Executive Summary

Healthcare Operations & Financial Intelligence Analysis

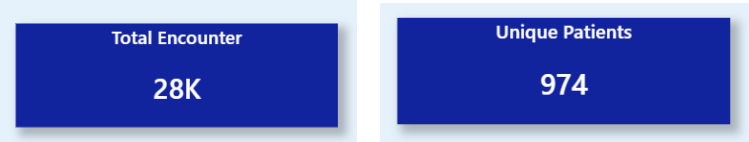
Project overview

This project examines healthcare operations using historical encounter, patient, & procedures datasets. The objective is to understand encounter patterns, healthcare costs, payer coverage, and patient activity to identify areas that may require operational improvement.

The analysis was conducted using Microsoft Power BI, with a focus on data cleaning, transformation, modelling, and visualization.

Key Findings

1. **Dataset Overview:** The encounters dataset contains 27,891 records, while the patients dataset contains 974 unique patient records.

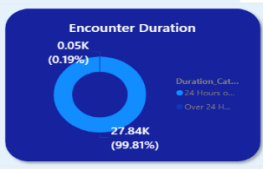


2. **Data quality:** No duplicate was encounter IDs or Patients IDs were identified, and all encounter patient references matched records in the patient dataset.

3. **Payer Coverage:** A total of 14K encounter had zero recorded payer coverage, representing approximately 48.7% of all encounters. This highlights the need to investigate how healthcare costs are covered and recorded.



4. **Encounter Duration:** 75 encounters lasted longer than 24 hours, representing approximately 0.27% of all encounters. These records may require further investigation to understand the nature of extended encounters.



5. **Missing Clinical Information:** 19,541 encounter records had missing reason codes, representing approximately **70.1% of the dataset**. This limits the ability to analyze encounter reasons comprehensively.

Recommendation

Based on the findings, the following actions are recommended:

- **Improve payer coverage monitoring:** Investigate encounters with zero recorded payer coverage to determine whether they represent self-funded services, uncovered expenses, or incomplete records.
- **Strengthen data documentation:** Improve the completeness of encounter reason codes to support more reliable reporting and operational analysis.
- **Review extended encounters:** Investigate encounters lasting more than 24 hours to understand their causes and identify potential operational bottlenecks.
- **Establish regular monitoring:** Use an interactive dashboard to monitor encounter volumes, costs, payer coverage, and patient activity over time.

Conclusion

The analysis provides an overview of healthcare operations and highlights important data quality and financial monitoring concerns. The high proportion of encounters with zero recorded payer coverage and missing reason codes presents an opportunity to improve financial visibility and data documentation.

Overall, this project demonstrates how data analytics can help healthcare organizations identify information gaps, monitor operational performance, and make more informed, evidence-based decisions.